ASSIGNMENT 3

Deadline: 11:59 pm, March 8, 2026

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1 Compulsory Questions

Question 1 (10%)

Please derive the closed-form solution for the weight vector $\mathbf{w}$ that minimizes the given regularized in-sample error. The regularized in-sample error is defined as:

$$E(\mathbf{w}) = \frac{1}{N}\sum_{n=1}^{N}\left(y_n - \mathbf{w}^T\mathbf{x}_n\right)^2 + \lambda\|\mathbf{w}\|^2,$$

where:

- $\mathbf{w} \in \mathbb{R}^d$ is the weight vector we aim to optimize,
- $\mathbf{x}_n \in \mathbb{R}^d$ is the input feature vector for the n-th training sample,
- $y_n \in \mathbb{R}$ is the corresponding target value for the n-th training sample,
- N is the total number of training samples,
- $\lambda \geq 0$ is the regularization parameter that controls the trade-off between the loss term and the regularization term,
- $\|\mathbf{w}\|^2 = \sum_{j=1}^{d} w_j^2$ is the squared L_2-norm of the weight vector $\mathbf{w}$,
- n indexes the data samples, where the total number of data samples is N.

Your task is to derive the closed-form solution for the weight vector $\mathbf{w}$ that minimizes $E(\mathbf{w})$. Provide the detailed steps involved in obtaining this solution.
Hint: Write down the matrix form of the data points and follow similar steps as on pages 84–85. Then, consider whether the matrix X^TX still needs to be invertible.

Question 2 (20%)

We sample three data points from an unknown binary classification target function:

$$(x_1, y_1) = (0.0, 1), \quad (x_2, y_2) = (1.0, -1), \quad (x_3, y_3) = (2.0, 1).$$

Here, $x_i \in \mathbb{R}$ is the input feature, and $y_i \in \{-1, +1\}$ is the corresponding label for the i-th data point. We use a one-dimensional logistic regression model to estimate the decision boundary:

$$P(y_i \mid x_i, \mathbf{w}) = \frac{1}{1 + e^{-y_i(w_1x_i + w_0)}},$$

where w_1 is the weight parameter for the input feature x_i, and w_0 is the one associated with the augmented data feature 1.

The log-likelihood loss function (negative log-likelihood) is:

$$L(\mathbf{w}) = -\frac{1}{n}\sum_{i=1}^{n}\ln\left(\frac{1}{1 + e^{-y_i(w_1x_i + w_0)}}\right).$$

(1) One-Step Gradient Descent Update Assume the initial parameters $w_1 = 0$, $w_0 = 0$ and a learning rate $\eta = 0.01$.

1. Compute the gradient of the loss function with respect to w_1 and w_0.
2. Perform one-step gradient descent to update w_1 and w_0.

(2) Loss Evaluation and Learning Rate Analysis Now assume the current parameters are still $w_1 = 0$, $w_0 = 0$, but set the learning rate to $\eta = 1.0$.

1. Compute the initial loss $L(\mathbf{w})$.
2. Perform one-step gradient descent with $\eta = 1.0$ and compute the new loss.
3. Discuss whether the learning rate is reasonable—does the loss decrease or increase?

Question 3 (20%)

In class, we derived the optimal weight vector $\mathbf{w}_{\text{lin}}$ by taking the gradient of the in-sample error and setting it to zero. Here, we will derive it using pure linear algebra (completing the square) without any calculus.

Assuming that $\mathbf{X}^T\mathbf{X}$ is invertible, show by direct comparison with the expanded definition of in-sample error $E_{\text{in}}(\mathbf{w}) = \frac{1}{N}(\mathbf{X}\mathbf{w} - \mathbf{y})^T(\mathbf{X}\mathbf{w} - \mathbf{y})$ that $E_{\text{in}}(\mathbf{w})$ can be rewritten as:

$$E_{\text{in}}(\mathbf{w}) = \frac{1}{N}\left((\mathbf{w} - (\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T\mathbf{y})^T(\mathbf{X}^T\mathbf{X})(\mathbf{w} - (\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T\mathbf{y}) + \mathbf{y}^T(\mathbf{I} - \mathbf{X}(\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T)\mathbf{y}\right)$$

Use this expression for $E_{\text{in}}(\mathbf{w})$ to obtain the optimal weight vector $\mathbf{w}_{\text{lin}}$. What is the minimum in-sample error? *[Hint: The matrix $\mathbf{X}^T\mathbf{X}$ is positive definite, which means $\mathbf{z}^T(\mathbf{X}^T\mathbf{X})\mathbf{z} \geq 0$ for any vector $\mathbf{z}$, and it equals 0 only if $\mathbf{z} = \mathbf{0}$.]*

Question 4 (20%)

Suppose we are given a dataset $\mathcal{D}$ with N training samples, where each sample has d features (plus the bias term $x_0 = 1$, so the total number of parameters is $d + 1$).

However, imagine a scenario where the features are completely useless for prediction. The true target y is generated *purely* by random noise:

$$y = \epsilon$$

where $\mathbb{E}[\epsilon] = 0$ and $\text{Var}(\epsilon) = \sigma^2$. This is equivalent to a true linear model $y = \mathbf{w}^{*T}\mathbf{x} + \epsilon$ where the true weight vector is strictly zero ($\mathbf{w}^* = \mathbf{0}$).

(a) Without doing any training, if we simply use the true model and guess $h(\mathbf{x}) = 0$ for all $\mathbf{x}$, what is the expected out-of-sample error E_{out}?

(b) Now, suppose we ignore the fact that the features are useless, and we forcefully apply Linear Regression to learn a weight vector $\mathbf{w}_{\text{lin}}$ from $\mathcal{D}$. Based on the theorems in the lecture slides, write down the expected in-sample error $\mathbb{E}[E_{\text{in}}(\mathbf{w}_{\text{lin}})]$ and the expected out-of-sample error $\mathbb{E}[E_{\text{out}}(\mathbf{w}_{\text{lin}})]$.

(c) Let's look at an extreme case. Suppose we collect exactly as many data points as the number of parameters, i.e., $N = d + 1$. Calculate the numerical values of $\mathbb{E}[E_{\text{in}}]$ and $\mathbb{E}[E_{\text{out}}]$ in this scenario.

(d) Briefly explain the phenomenon observed in part (c). Geometrically or algebraically, why is E_{in} so low? Why is E_{out} so high compared to part (a)? What does this tell us about the risk of using high-dimensional linear models ($d \approx N$)? *[Hint: Detailed analysis of this phenomenon is discussed in Question 9.]*

Question 5 (30%)

Programming Problem. Consider a linear model for generating data:

$$y = w_1 x + w_0 + \epsilon,$$

where: x is uniformly distributed in the interval $[0, 10]$, ϵ is independent and identically distributed (iid) Gaussian noise with mean 0 and variance $\sigma^2 = 1$, and $w_1 = 2.0$ and $w_0 = 1.0$ are the true parameters of the linear model.

Part 1: Linear Regression with iid Noise

1. Generate a dataset with $N = 100$ samples by sampling x uniformly from $[0, 10]$ and computing y using the linear model with iid Gaussian noise. (The data is already generated with data `x_q1.txt` and data `y_q1.txt`, please use directly.)
2. Perform linear regression using the theoretical equation:

$$\mathbf{w} = (X^\top X)^{-1}X^\top y,$$

where X is the design matrix (with a column of ones for the bias term) and y is the target vector.

3. Compute the residual error for the theoretical solution, defined as:

$$\text{Residual Error} = \frac{1}{N}\sum_{i=1}^{N}(y_i - \hat{y}_i)^2,$$

where $\hat{y}_i = w_1 x_i + w_0$ is the predicted value.

4. Use gradient descent to solve the linear regression problem. The loss function is:

$$L(\mathbf{w}) = \frac{1}{2N}\sum_{i=1}^{N}(y_i - (w_1 x_i + w_0))^2.$$

The gradient descent updates are:

$$w_1^{\text{new}} = w_1 - \eta\frac{\partial L}{\partial w_1}, \qquad w_0^{\text{new}} = w_0 - \eta\frac{\partial L}{\partial w_0},$$

where $\eta = 0.01$ is the learning rate. Stop the algorithm when the Euclidean norm of the gradient is smaller than 0.0001.

5. Plot the residual errors for gradient descent at each iteration and compare the final residual error with the theoretical result.

Part 2: Linear Regression with Correlated Noise Now consider the case where the noise ϵ is non-iid and correlated. Generate the noise as follows:

$$\epsilon_i = 0.5\epsilon_{i-1} + \nu_i,$$

where ν_i is iid Gaussian noise with mean 0 and variance $\sigma^2 = 1$, and $\epsilon_1 = \nu_1$.

1. Generate a new dataset with $N = 100$ samples using the same linear model but with correlated noise. (The data is already generated with data `x_q2.txt` and data `y_q2.txt`, please use directly.)

2. Repeat steps 2–5 from Part 1 using this new dataset.

3. Compare the residual errors and convergence behavior of gradient descent for the iid and correlated noise cases.

Question 1

$$E(\vec{w}) = \frac{1}{N}\sum_{n=1}^{N}(y_n - \vec{w}^T\vec{x}_n)^2 + \lambda\|\vec{w}\|^2$$

Let $X = \begin{bmatrix} \vec{x}_1^T \\ \vec{x}_2^T \\ \vdots \\ \vec{x}_N^T \end{bmatrix}$, $\vec{y} = \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_N \end{bmatrix}$, then $X\vec{w} = \begin{bmatrix} \vec{w}^T\vec{x}_1 \\ \vec{w}^T\vec{x}_2 \\ \vdots \\ \vec{w}^T\vec{x}_N \end{bmatrix}$

$$E(\vec{w}) = \frac{1}{N}(X\vec{w} - \vec{y})^T(X\vec{w} - \vec{y}) + \lambda\vec{w}^T\vec{w}$$

$$= \frac{1}{N}(\vec{w}^T X^T X\vec{w} - 2\vec{y}^T X\vec{w} + \vec{y}^T\vec{y}) + \lambda\vec{w}^T\vec{w}$$

$$\nabla_{\vec{w}}E(\vec{w}) = \frac{2}{N}X^T X\vec{w} - \frac{2}{N}X^T\vec{y} + 2\lambda\vec{w}$$

For $\nabla_{\vec{w}}E(\vec{w}) = 0$,

$$\frac{2}{N}X^T X\vec{w} - \frac{2}{N}X^T\vec{y} + 2\lambda\vec{w} = 0$$

$$\frac{2}{N}X^T X\vec{w} + 2\lambda\vec{w} = \frac{2}{N}X^T\vec{y}$$

$$(X^T X + N\lambda I)\vec{w} = X^T y$$

$$\vec{w} = (X^T X + N\lambda I)^{-1}X^T\vec{y}$$

Question 2

(1)(1) $w_1 = 0$, $w_0 = 0$, $\eta = 0.01$

$$\frac{\partial L}{\partial w_0} = -\frac{1}{n}\sum_{i=1}^{n} \frac{y_i}{1+e^{y_i(w_1 x_i + w_0)}}$$

$$\frac{\partial L}{\partial w_1} = -\frac{1}{n}\sum_{i=1}^{n} \frac{y_i x_i}{1+e^{y_i(w_1 x_i + w_0)}}$$

$\because$ $w_1 = 0$, $w_0 = 0$

$$\therefore \frac{\partial L}{\partial w_0} = -\frac{1}{n}\sum_{i=1}^{n} \frac{y_i}{2}$$

$$\frac{\partial L}{\partial w_1} = -\frac{1}{n}\sum_{i=1}^{n} \frac{y_i x_i}{2}$$

$$\frac{\partial L}{\partial w_0} = -\frac{1}{3}\left(\frac{1}{2} - \frac{1}{2} + \frac{1}{2}\right)$$

$$= -\frac{1}{6}$$

$$\frac{\partial L}{\partial w_1} = -\frac{1}{3}\left(\frac{0(1)}{2} + \frac{1(-1)}{2} + \frac{2(1)}{2}\right)$$

$$= -\frac{1}{3}\left(\frac{1}{2}\right)$$

$$= -\frac{1}{6}$$

(2) $w_0' = 0 - 0.01\left(-\frac{1}{6}\right)$

$$= \frac{1}{600} = 0.00167$$

$$w_1' = 0 - 0.01\left(-\frac{1}{6}\right) = \frac{1}{600} = 0.00167$$

Question 2

(2) (1) $w_1 = 0$, $w_0 = 0$, $\eta = 1$

$$L(0,0) = \frac{1}{3}\left(\ln(1+e^0) + \ln(1+e^0) + \ln(1+e^0)\right)$$

$$= \frac{1}{3}(3\ln 2) = \ln 2 = 0.6931$$

(2) $w_0'' = 0 - 1\left(-\frac{1}{6}\right) = \frac{1}{6}$

$w_1'' = 0 - 1\left(-\frac{1}{6}\right) = \frac{1}{6}$

$$L(w'') = \frac{\ln(1+e^{-(1)(0+\frac{1}{6})}) + \ln(1+e^{-(-1)(\frac{1}{6}+\frac{1}{6})}) + \ln(1+e^{-(1)(\frac{1}{6}\cdot 2+\frac{1}{6})})}{3}$$

$$= \frac{\ln(1+e^{-\frac{1}{6}}) + \ln(1+e^{\frac{1}{3}}) + \ln(1+e^{-\frac{1}{2}})}{3}$$

$$= 0.6533$$

(3) $\because$ $0.6533 < 0.693$, $\therefore$ The loss decreases and therefore it is reasonable. If η is too large, it may oscillate or increase.

Question 3

$X^T X$ is invertible

$$E_{in}(\vec{w}) = \frac{1}{N}(X\vec{w} - \vec{y})^T (X\vec{w} - \vec{y})$$
$$= \frac{1}{N}(\vec{w}^T X^T X \vec{w} - 2\vec{y}^T X \vec{w}) + \vec{y}^T y)$$

By linear regression, $\vec{w}_{lin} = (X^T X)^{-1} X^T \vec{y}$

$$E_{in}(\vec{w}) = \frac{1}{N}\Big((\vec{w} - (X^T X)^{-1}(X^T \vec{y})^T (X^T X)(\vec{w} - (X^T X)^{-1}(X^T \vec{y})$$
$$- (X^T \vec{y})(X^T X)^{-1}(X^T \vec{y}) + y^T y)$$
$$= \frac{1}{N}\Big((\vec{w} - \vec{w}_{lin})^T X^T X (w - w_{lin}) + y^T (I - X(X^T X)^{-1} X^T) y\Big)$$

E_{in} will be min. when $w = w_{lin}$

$$\therefore E_{in\text{-}min} = \frac{1}{N} y^T (I - X(X^T X)^{-1} X^T) y$$

Question 4

(a) Guess $h(x) = 0 \ \forall X$

$$E_{out} = E[(h(x) - y)^2)]$$
$$= E[y^2] = E[\varepsilon^2]$$

$\because Var(\varepsilon) = \sigma^2$

$\therefore E_{out} = \sigma^2$

(b) linear regression $= \vec{w}_{lin} = (X^T X)^{-1} X^T \vec{y}$

$$\mathbb{E}[E_{in}(\vec{w}_{lin})] = \sigma^2 (1 - \frac{d+1}{N})$$

$$\mathbb{E}[E_{out}(\vec{w}_{lin})] = \sigma^2 (1 + \frac{d+1}{N})$$

(c) $N = d+1$

$$\mathbb{E}[E_{in}] = \sigma^2 (1 - \frac{d+1}{d+1})$$
$$= 0$$

$$\mathbb{E}(E_{out}) = \sigma^2 (1 + \frac{d+1}{d+1})$$
$$= 2\sigma^2$$

(d) When $N = d+1$, the model has enough parameters to fit the noice.

$\therefore E_{in} = 0$ but it remembers the noise which leads $E_{out} = 2\sigma^2 > \sigma^2$.

$\therefore$ When $d \approx N$, using high-dimensional model will overfit.

Question 5

```
--- Part 1: iid Noise ---
Theoretical Weights (w0, w1): [1.22215108 1.9936935 ]
Theoretical Residual Error: 0.992439
GD Final Weights (w0, w1): [1.22177955 1.99375128]
GD Final Residual Error: 0.992439
Successfully saved plot to: residual_error_q1.png

--- Part 2: Correlated Noise ---
Theoretical Weights (w0, w1): [1.21596141 2.03668315]
Theoretical Residual Error: 1.206061
GD Final Weights (w0, w1): [1.21558926 2.03674102]
GD Final Residual Error: 1.206061
Successfully saved plot to: residual_error_q2.png
```

(1) Theoretical residual error $= 0.992439$

Final residual error $= 0.992439$

They are the same, meaning that GD has converged to the global minimum of the convex loss function. It shows GD is effective for linear regression under iid noise assumption.

(2) Residual error for correlated case $= 1.206061 > 0.992439$.

This increase is because of the autoregressive noise violates the iid assumption.

Besides, both cases show similar convergence behavior, they have a sharp drop in residual error for first 100 iteration, followed by stabilization.

Attachment

Declaration for written assignment

I am/we are submitting the assignment for:
☑ an individual project or
□ a group project on behalf of all members of the group. It is hereby confirmed that the submission is authorized by all members of the group, and all members of the group are required to sign this declaration.

I/We declare that: (i) the assignment here submitted is original except for source material explicitly acknowledged/all members of the group have read and checked that all parts of the piece of work, irrespective of whether they are contributed by individual members or all members as a group, here submitted are original except for source material explicitly acknowledged; (ii) the piece of work, or a part of the piece of work has not been submitted for more than one purpose (e.g. to satisfy the requirements in two different courses) without declaration; and (iii) the submitted soft copy with details listed in the <Submission Details> is identical to the hard copy(ies), if any, which has(have) been / is(are) going to be submitted. I/We also acknowledge that I am/we are aware of the University's policy and regulations on honesty in academic work, and of the disciplinary guidelines and procedures applicable to breaches of such policy and regulations, as contained in the University website http://www.cuhk.edu.hk/policy/academichonesty/.

In the case of a group project, we are aware that all members of the group should be held responsible and liable to disciplinary actions, irrespective of whether he/she has signed the declaration and whether he/she has contributed, directly or indirectly, to the problematic content.

I/we declare that I/we have not distributed/ shared/ copied any teaching materials without the consent of the course teacher(s) to gain unfair academic advantage in the assignment/ course.

I/we declare that I/we have read and understood the University's policy on the use of AI for academic work. I/we confirm that I/we have complied with the instructions given by my/our course teacher(s) regarding the use of AI tools for this assignment and consent to the use of AI content detection software to review my/our submission.

I/We also understand that assignments without a properly signed declaration by the student concerned and in the case of a group project, by all members of the group concerned, will not be graded by the teacher(s).

Chena	8 / 3 / 2025
Signature(s)	Date
Fung Cheuk Nam	1155192095
Name~~(s)~~	Student ID(s)
CSCI 3320	Fundamental of Machine Learning
Course code	Course title

Use of AI tools.

I acknowledge the use of Grok, ChatGPT, Gemini (2026, Mar 8)

for content guidance, code guidance